A Call for an AI Network for HEP(N) Theory

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The Call: An AI-oriented network for HEP(N) theorists

- New AI tools are changing physics research
- The tools we need are not cheap.
- Science funding has moved to AI.
- Experimental HEP is already organizing.

Unless we do too, we do not get
the frontier resources we need.

What changed?

- AI capabilities
- Access to good and frontier models
- Shrinking of usual funding sources
- Refocus of science funding into AI

The Genesis Mission and its Platform

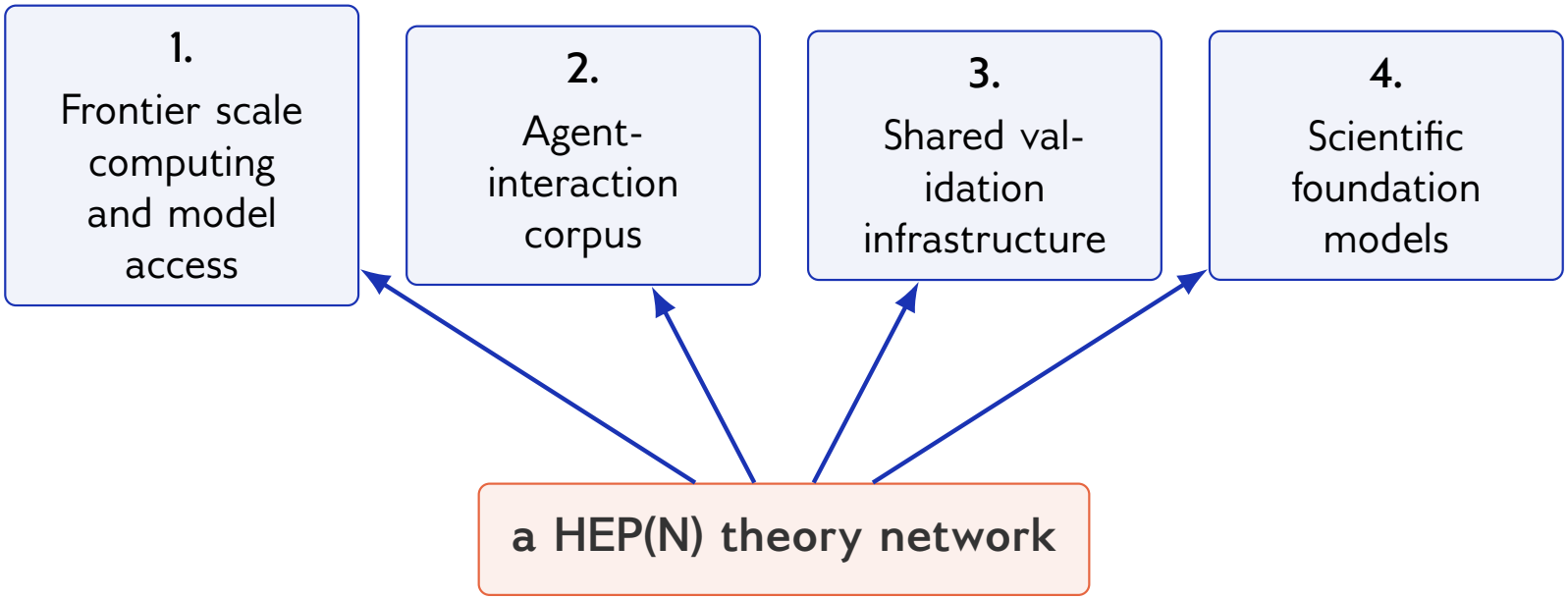
Why organize? New technologies mean new capabilities

The methods of particle theory can change, but the practice does not

- What we do can be the same:
  - build models, compute consequences, compare to data
  - study data, infer models, calculate new consequences
- How we do can it change:
  - search literature, make derivations, write code, run calculations, summarize results, (insert your least favorite but necessary scientific task): an agent can do part of each
- What we do can also change:
  - Theory of AI
  - Discover new representations of the natural world
  - *etc.*

Potential assets only a community can have

Out of reach for one PI, one group, most single institutions



These assets did not exist five years ago.
Not one of them can be obtained by a single researcher or research group.

1. Frontier scale computing and model access

Beyond a single PI's budget, and beyond most universities'

- Frontier models, and enough tokens to do real work with them.
- Computing resources for training (GPUs) and for building foundation models.
- A fair share of both, decided by the community, rather than by which group happened to write which grant.

2. Agent-interaction corpus

The record of scientists working with agents is itself data

- Long sessions, wrong turns, corrections, and the reasoning that got there.
- This is training and evaluation material for scientific agents.
- Pooled across many groups, it can improve our models and interactions with them.

3. Shared validation infrastructure

Storage and access of benchmarks, and tools for fair comparison

- A benchmark held privately is worth very little. Held in common it settles arguments.
- Standardization, so two groups' claims can be put side by side.
- Variety, so a benchmark measures physics and not one group's taste in problems.
- Standardization and variety together is something only a community can give.

4. Scientific foundation models

Multi-modal representations of experiment AND theory data

- Building, storage and access: again, no single group can do this.
- Experiment data is half the picture. Theory data, simulated and derived, is the other half, and it is ours to provide.
- Value scales with the diversity of the data. Which is an argument for breadth, and so for organizing.

The experiments are already organizing

- The HEP experimental community has been meeting on exactly this: the Blueprint process in DC, Clariphy, and the rest.
- Their list is reconstruction, trigger, simulation, analysis. That is the right list for them.
- Two wings of one national HEP AI effort, not two parallel enterprises.
- But they are not thinking about us, and nobody is going to write down theory's needs on our behalf.

Does Theory have a special place?

Some scientific tasks are just common with experiment

- Optimization and efficiency problems likely share common solutions.
- Those are wins for all, and the easiest wins to argue for.
- But if we stop there, theory is treated as a small experiment, and funded like one.

What theory does that nobody else does

- Theorists generate simulation data, for experiment and for themselves.
- And other sorts of data:
  - equations (trial truth statements)
  - functions
- Theory behind NN itself: connections to QFT and UQ.
- Reasoning trail. Logic: training data.
- Big problems that need to be studied, to guide current research and to decide on new experiments.

Phenomenology techniques and tools used HEP-wide have come from the theory/pheno community.

Theorists work differently

An allocation model built around experiment will not fit us

- Rubber ducking.
- Long, interactive sessions.
- “Failures” can be good data.

Resources should not be tied to how the experimental community operates.

Why organize? Survive

We live in an age where value is measured by profit and efficiency

- The declining HEP budget has been sustained by finding clever ways to advance both science and computing and software at once. That is the sweet spot, and it is why the money kept coming.
- Funding emphasis is now on AI.
- Pure theory funding alone will not carry us. We need to find the same sweet spot again.

Why organize? Thrive

Survival is the floor. These are the reasons to want it.

1. Deeper dive into data. Questions we currently do not ask because they cost too many person-hours.

2. Monte Carlo of ideas. Generate and test model space the way we already generate and test event space.

3. Drawing in the best and brightest talent. The students who want to work this way will go where they can.

What would a theory community even mean?

- access to the same frontier and other models
- fair share of tokens
- shared benchmarks (agreements)
- frontier computing for training custom models and building foundation models
- shared foundation models
- corpus of scientist-agent interactions
- plugins to share innovative tools

Organizing does not cost you your independence

Unchanged

- ability to explore **your own ideas** with some intellectual control

Changes

- working in **larger groups**
- working on **community service projects**
- the **ASCR model**

Community testimonial

First, I am on an experiment

- That has helped me discover problems, and find solutions, that I would have missed. I have learned and I have taught.

- **Claim:** event generator groups are theory's closest equivalent to an experimental collaboration.

Event generator developers model

Shared assets

- git repos
  - integration of work
  - builds and validation
  - issue tracking
- purpose and vision
- sustainability plan
- policies

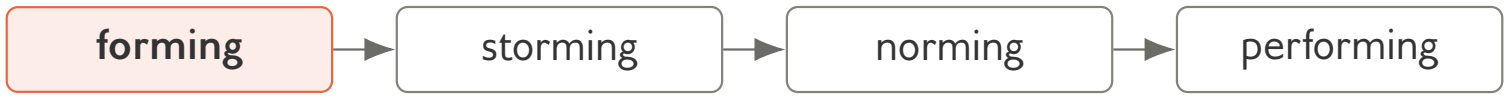
With

- local research interests
- projects that can be attached to students and grants

It works! USQCD is a larger community effort that has done exactly this (and more) for lattice theory for over two decades.

Where we are

The natural sequence: forming, storming, norming, performing



now

Now is the time for forming.

SAINT

Self-organizing

AI Network

for Theorists

(Self-organizing, shared, or any good word beginning with S.)

Where to start

Three steps, which need little or no new money to begin

1. National Lab theory groups serve as an anchor: organize a meeting, within the year, focused on the four assets.

2. A pilot agent-interaction repo on the GMP, open to all registering theorists. This one first: it is the cheapest to build, and its value grows fastest with the number of groups contributing.

3. Connect to the broader national effort through the GMP, using common formats.

And throughout: education and workforce development.

The bottom line

Theorists can keep what is valuable in how they work and still share resources

- Organizing is how we survive the shift in funding, and how we thrive on what the tools now make possible.
- Event generator groups and USQCD show it already works, in our own community.
- Theory needs a voice in a field-wide AI effort, and that requires a community.

Now is the time for forming. Start with one meeting and one repo.

Blueprint White Paper

Blueprint for a Particle Theory AI Community for AI-Enabled Discoveries

George T. Fleming[1], Daniel C. Hackett[1], James Halverson[2], T. J. Hobbs[3], Stefan Hoeche[1], Brandon Kriesten[3], Konstantin T. Matchev[4], Tony Menzo[1,4], Stephen Mrenna[1], Benjamin Nachman[5], Prasanth Shyamsundar[1], David Shih[6], Jesse Thaler[7], Jure Zupan[8]

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Abstract

Rapid advances in the capabilities and reach of artificial intelligence (AI) are changing research in particle physics. In this new era, AI is an innovative tool for theoretical physicists, while the interaction of AI and particle theory is itself a developing field. We argue that particle theory requires a community-scale effort to realize its full potential. We show how particle theory occupies a unique position in the AI for physics landscape, and describe the complementarity of AI and theory developments. We conclude with a series of recommendations on how to build an AI theory community for the Genesis Mission.

http://smrenna.github.io/share/whitepaper.pdf

Link for gathering endorsements later this week.